

Exercise 4.4

1. Find the square roots of the following polynomials by factorization method:

(i) $x^2 - 8x + 16$

$$x^2 - 8x + 16 = (x)^2 - 2(x)(4) + (4)^2$$

$$x^2 - 8x + 16 = (x - 4)^2$$

Taking square root on both sides

$$\sqrt{x^2 - 8x + 16} = \sqrt{(x - 4)^2}$$

$$\sqrt{x^2 - 8x + 16} = \pm(x - 4)$$

(ii) $9x^2 + 12x + 4$

$$9x^2 + 12x + 4 = (3x)^2 + 2(3x)(2) + (2)^2$$

$$9x^2 + 12x + 4 = (3x + 2)^2$$

Taking square root on both sides

$$\sqrt{9x^2 + 12x + 4} = \sqrt{(3x + 2)^2}$$

$$\sqrt{9x^2 + 12x + 4} = \pm(3x + 2)$$

(iii) $36a^2 + 84a + 49$

$$36a^2 + 84a + 49 = (6a)^2 + 2(6a)(7) + (7)^2$$

$$36a^2 + 84a + 49 = (6a + 7)^2$$

Taking square root on both sides

$$\sqrt{36a^2 + 84a + 49} = \sqrt{(6a + 7)^2}$$

$$\sqrt{36a^2 + 84a + 49} = \pm(6a + 7)$$

(iv) $64y^2 - 32y + 4$

$$64y^2 - 32y + 4 = (8y)^2 - 2(8y)(2) + (2)^2$$

$$64y^2 - 32y + 4 = (8y - 2)^2$$

Taking square root on both sides

$$\sqrt{64y^2 - 32y + 4} = \sqrt{(8y - 2)^2}$$

$$\sqrt{64y^2 - 32y + 4} = \pm(8y - 2)$$

(v) $200t^2 - 120t + 18$

$$200t^2 - 120t + 18 = 2[100t^2 - 60t + 9]$$

$$200t^2 - 120t + 18 = 2[(10t)^2 - 2(10t)(3) + (3)^2]$$

$$200t^2 - 120t + 18 = 2(10t - 3)^2$$

Taking square root on both sides

$$\sqrt{200t^2 - 120t + 18} = \sqrt{2(10t - 3)^2}$$

$$\sqrt{200t^2 - 120t + 18} = \pm\sqrt{2}(10t - 3)$$

(vi) $40x^2 + 120x + 90$

$$40x^2 + 120x + 90 = 10[4x^2 + 12x + 9]$$

$$40x^2 + 120x + 90 = 10[(2x)^2 + 2(2x)(3) + (3)^2]$$

$$40x^2 + 120x + 90 = 10(2x + 3)^2$$

Taking square root on both sides

$$\sqrt{40x^2 + 120x + 90} = \sqrt{10(2x + 3)^2}$$

$$\sqrt{40x^2 + 120x + 90} = \pm\sqrt{10}(2x + 3)$$

2. Find the square root of the following polynomials by division method:

(i) $4x^4 - 28x^3 + 37x^2 + 42x + 9$

$$\begin{array}{r}
 2x^2 - 7x - 3 \\
 2x^2 \overline{) 4x^4 - 28x^3 + 37x^2 + 42x + 9} \\
 \underline{\pm 4x^4} \\
 4x^2 - 7x \\
 \underline{ \mp 28x^3 \pm 49x^2} \\
 4x^2 - 14x - 3 \\
 \underline{ \mp 12x^2 \pm 42x \pm 9} \\
 0
 \end{array}$$

Square Root = $\pm(2x^2 - 7x - 3)$

(ii) $121x^4 - 198x^3 - 183x^2 + 216x + 144$

$$\begin{array}{r}
 11x^2 - 9x - 12 \\
 11x^2 \overline{) 121x^4 - 198x^3 - 183x^2 + 216x + 144} \\
 \underline{\pm 121x^4} \\
 22x^2 - 9x \\
 \underline{ \mp 198x^3 \pm 81x^2} \\
 22x^2 - 18x - 12 \\
 \underline{ \mp 264x^2 \pm 216x \pm 144} \\
 0
 \end{array}$$

Square Root = $\pm(11x^2 - 9x - 12)$

(iii) $x^4 - 10x^3y + 27x^2y^2 - 10xy^3 + y^4$

$$\begin{array}{r}
 x^2 - 5xy + y^2 \\
 x^2 \overline{) x^4 - 10x^3y + 27x^2y^2 - 10xy^3 + y^4} \\
 \underline{\pm x^4} \\
 2x^2 - 5xy \\
 \underline{ \mp 10x^3y \pm 25x^2y^2} \\
 2x^2 - 10xy + y^2 \\
 \underline{ \pm 2x^2y^2 \mp 10xy^3 \pm y^4} \\
 0
 \end{array}$$

Square Root = $\pm(x^2 - 5xy + y^2)$

(iv) $4x^4 - 12x^3 + 37x^2 - 42x + 49$

$$\begin{array}{r}
 2x^2 - 3x + 7 \\
 2x^2 \overline{) 4x^4 - 12x^3 + 37x^2 - 42x + 49} \\
 \underline{\pm 4x^4} \\
 4x^2 - 3x \\
 \underline{ \mp 12x^3 \pm 9x^2} \\
 4x^2 - 6x + 7 \\
 \underline{ \pm 12x^2 \mp 42x \pm 49} \\
 0
 \end{array}$$

Square Root = $\pm(2x^2 - 3x + 7)$

3. An investor's return $R(x)$ in rupees after investing x thousand rupees is given by quadratic expression: $R(x) = -x^2 + 6x - 8$ Factorize the expression and find the investment levels that result in zero return.

$$\begin{aligned}
 R(x) &= -x^2 + 6x - 8 \\
 R(x) &= -[x^2 - 6x + 8] \\
 R(x) &= -[x^2 - 4x - 2x + 8] \\
 R(x) &= -[x(x - 4) - 2(x - 4)] \\
 R(x) &= -[(x - 2)(x - 4)] \\
 R(x) &= -(x - 2)(x - 4)
 \end{aligned}$$

Put $R(x) = 0$ for zero return:

$$R(x) = 0$$

$$-(x - 2)(x - 4) = 0$$

$$\begin{array}{ccc} \text{Either} & x - 2 = 0 & \text{or} & x - 4 = 0 \\ & x = 2 & , & x = 4 \end{array}$$

Since x is in thousands of rupees, the investment levels that result in zero return are **Rs. 2000** and **Rs. 4000**.

4. A company's profit $P(x)$ in rupees from selling x units of a product is modeled by the cubic expression: $P(x) = x^3 - 15x^2 + 75x - 125$ Find the break-even point(s), where the profit is zero.

$$P(x) = x^3 - 15x^2 + 75x - 125$$

$$P(x) = (x)^3 - 3(x)^2(5) + 3(x)(5)^2 - (5)^3$$

$$P(x) = (x - 5)^3$$

Put $P(x) = 0$ for break-even point(s)

$$P(x) = 0$$

$$(x - 5)^3 = 0$$

Taking cube root on both sides

$$\sqrt[3]{(x - 5)^3} = \sqrt[3]{0}$$

$$x - 5 = 0$$

$$x = 5$$

Thus, the break-even point is at **5 units**.

5. The potential energy $V(x)$ in an electric field varies as a cubic function of distance x , given by: $V(x) = 2x^3 - 6x^2 + 4x$ Determine where the potential energy is zero.

$$V(x) = 2x^3 - 6x^2 + 4x$$

$$V(x) = 2x[x^2 - 3x + 2]$$

$$V(x) = 2x[x^2 - 2x - x + 2]$$

$$V(x) = 2x[x(x - 2) - 1(x - 2)]$$

$$V(x) = 2x[(x - 2)(x - 1)]$$

$$V(x) = 2x(x - 2)(x - 1)$$

Put $V(x) = 0$ for zero potential energy

$$\begin{array}{ccccccc} & & & V(x) = 0 & & & \\ & & & 2x(x - 2)(x - 1) = 0 & & & \\ \Rightarrow & 2x = 0 & , & x - 2 = 0 & , & x - 1 = 0 & \\ & x = 0 & , & x = 2 & , & x = 1 & \end{array}$$

Thus, the potential energy is zero at $x = 0$, $x = 1$, and $x = 2$.

6. In structural engineering, the deflection $Y(x)$ of a beam is given by: $Y(x) = 2x^2 - 8x + 6$. This equation gives the vertical deflection at any point x along the beam. Find the points of zero deflection.

$$Y(x) = 2x^2 - 8x + 6$$

$$Y(x) = 2[x^2 - 4x + 3]$$

$$Y(x) = 2[x^2 - 3x - x + 3]$$

$$Y(x) = 2[x(x - 3) - 1(x - 3)]$$

$$Y(x) = 2[(x - 3)(x - 1)]$$

$$Y(x) = 2(x - 3)(x - 1)$$

Put $Y(x) = 0$ for points of zero deflection

$$Y(x) = 0$$

$$2(x - 3)(x - 1) = 0$$

Since $2 \neq 0$, so

Either

$$\begin{aligned}x - 3 &= 0 \\x &= 3\end{aligned}$$

or

$$\begin{aligned}x - 1 &= 0 \\x &= 1\end{aligned}$$

Therefore, the points of zero deflection are at $x = 1$ and $x = 3$.

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